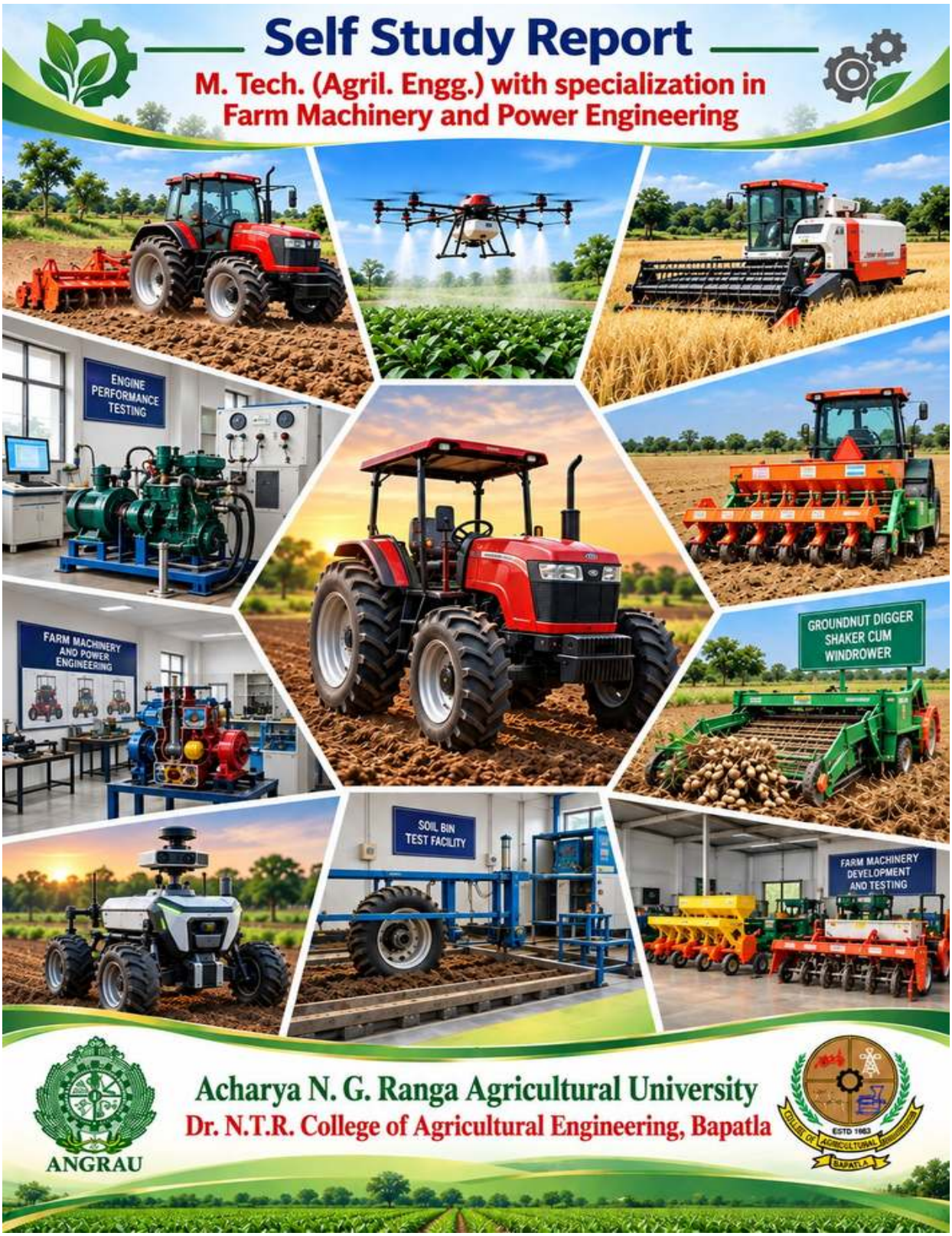

Self Study Report
M. Tech. (Agril. Engg.) with specialization in
Farm Machinery and Power Engineering
ENGINE PERFORMANCE TESTING
FARM MACHINERY AND POWER ENGINEERING
GROUNDNUT DIGGER SHAKER CUM WINDROWER
SOIL BIN TEST FACILITY
FARM MACHINERY DEVELOPMENT AND TESTING
ANGRAU
Acharya N. G. Ranga Agricultural University
Dr. N.T.R. College of Agricultural Engineering, Bapatla
ESTD 1963
BAPATLA

SELF STUDY REPORT
(2025-26)



Submitted to

ICAR Accreditation Peer Review Team

For Accreditation of

M. Tech. (Agril. Engg.) with specialization in Farm Machinery and Power Engineering

Submitted by

Dr. N.T.R. College of Agricultural Engineering, Bapatla – 522 101, Andhra Pradesh

INDEX

S. No.	Self Study Report	Page No.
1	M. Tech. (Farm Machinery and Power Engineering)	4-49

ACCREDITATION REPORT – TASK COMMITTEE

Chairman	:	Dr. M. Sardar Baig, Associate Dean
Coordinator	:	Dr. K.V.S. Rami Reddy, Professor and Head, Dept. of FMPE
Members	:	Dr. D. Vijay Kumar, Assistant Professor, Dept. of FMPE
		Dr. A. Srinivasa Rao, Assistant Professor, Dept. of FMPE
		Dr. B. Raj kiran, Assistant Professor, Dept. of FMPE
		Dr. A. Ashok Kumar, Assistant Professor, Dept. of FMPE
		Er. N. Kishore, Assistant Professor, Dept. of FMPE
		Dr. G. Veera Prasad, Scientist, AICRP on FIM

Contents

	M.Tech. (Farm Machinery and Power Engineering)	
6.4.1	Brief History of the Degree Programme	**4**
6.4.2	Faculty Strength	**5**
6.4.3	Technical and Supporting Staff	**7**
6.4.4	Classrooms and Laboratories	**8**
6.4.5	Conduct of Practical and Hands-on Training	**14**
6.4.6	Supervision of Students in PG Programmes	**16**
6.4.7	Feedback of Stakeholders (Students, Parents, Industries, Employers, Farmers, etc.)	**18**
6.4.8	Student Intake and Attrition in the Programme for Last Five Years	**32**
6.4.9	ICT Application in Curricula Delivery	**33**
6.4.11	Certificate	**51**

SELF STUDY REPORT FOR ACCREDITATION OF PROGRAMME
M.Tech. (Farm Machinery and Power Engineering)
Dr. N.T.R. College of Agricultural Engineering, Bapatla

6.4.1 Brief History of the Degree Programme

B.Tech. (Agricultural Engineering) undergraduate (UG) degree programme was started in August 1983 with an intake of 20 students in the Department of Agricultural Engineering, Agricultural College, Bapatla. From the next year onwards, the seats were increased to 30. The UG programme was upgraded to the College of Agricultural Engineering (CAE), Bapatla, in 1990 based on the recommendations of a high-level committee appointed by the Government of Andhra Pradesh. It is the first college of agricultural engineering in Acharya N.G. Ranga Agricultural University. It was started with the aim of imparting knowledge and skills to the students in the disciplines of agricultural engineering education to serve the farming community and the rural people of the combined state of Andhra Pradesh and to make agriculture more remunerative to the farmers by bringing in the engineering interventions in agriculture. The College was moved to its new premises in May 1994. Since then, the College expanded rapidly both academically and physically. Many infrastructural facilities were added in terms of buildings and equipment for improving academic status in agricultural engineering. This college was accredited by Indian Council of Agricultural Research, New Delhi during 1999-2000 in the first phase itself based on the ICAR's Review Team Report and during the subsequent phases also. The latest accreditation of the ICAR for this College was in 2019-20. The College was under the faculty of Agriculture until 2010 when a new Faculty of Agricultural Engineering and Technology was established by the University as per the provisions of the University Act. The College was renamed as Dr. N.T.R. College of Agricultural Engineering (Dr NTR CAE) in the year 2018 in memory of Late Dr. N.T. Rama Rao, the then Chief Minister of combined Andhra Pradesh and who was the person who laid foundation stone for the College building on 24-02-1988. Today, Dr. N.T.R. CAE is widely recognized for its quality education; need based research and transfer of technology in service of farming community.

M.Tech. (Farm Machinery and Power Engineering) was started from the academic year 2006-07 onwards at Dr. NTR College Agricultural Engineering, Bapatla. Students of M.Tech.

(Farm Machinery and Power Engineering) are provided with a monthly stipend of Rs.7,000/- by Acharya N.G. Ranga Agricultural University. ICAR specified syllabus is being followed for teaching M.Tech. (Farm Machinery and Power Engineering) courses as per BSMA guidelines.

The College has state-of-the-art infrastructure and facilities and the academic environment which combines academic rigor with a high level of interaction with government departments in serving the farming community of the nation in general and the present state of Andhra Pradesh after bifurcation in particular. The driving force behind the high level of academic excellence and quality education is the dedicated team of over twenty core faculty drawn from different specializations.

VISION

To make Dr. N.T.R. College of Agricultural Engineering, Bapatla an integrated center for academic excellence in agricultural engineering education known for its contribution to agriculture and allied fields in making agriculture more remunerative to farmers.

MISSION

To serve the farming community and enhance their quality of life through excellence in education, research, development and transfer of agricultural technologies.

6.4.2 Faculty Strength

There is enough faculty available to teach all the courses of the PG program of the department. The faculty to teach B. Tech. (Agricultural Engineering) degree is also available to teach PG courses. College is having sanctioned a strength of 31 faculty positions. Apart from 16 filled in posts of the College, there are 6 Teaching Associates working in the College, a Scientist of AICRP on Farm Implements Machinery Scheme and support from Agricultural College to cater the needs of the academic requirement. The College meets the number of the faculty recommended by the Indian Council of Agricultural Research, New Delhi.

Table 1. Sanctioned and In-Position Faculty at Dr. NTR CAE, Bapatla

SI. No.	Position	Sanctioned Faculty	Faculty in Place	Vacant Positions
1	Associate Dean	1	1	0
2	Professor	5	5+2*	0
3	Associate Professor	8	2+2*	4
4	Assistant Professor	17	8+16*	0
	Total	**31**	**36 (16 + 20*)**	**4**

** Staff from Agricultural College/ Research/Extension and Contractual faculty*

Table 2. Department-wise Sanctioned and In-Position Faculty at Dr. NTR CAE, Bapatla

Sl. No.	Department	Assistant Professor		Associate Professor		Professor		Total	
	Faculty / Department	**SS**	**IP**	**SS**	**IP**	**SS**	**IP**	**SS**	**IP**
1	Associate Dean	-	-	-	-	1	1	**1**	**1**
2	Farm Machinery and Power Engineering	5	5+1*	2	-	1	1	**8**	**7**
3	Processing and Food Engineering	3	3*	2	1+1*	1	2	**6**	**7**
4	Irrigation and Drainage Engineering	2	2+1*	1	1	1	1	**4**	**5**
5	Soil and Water Conservation Engineering	2	1+2*	1	-	1	1	**4**	**4**
6	Renewable Energy Engineering	1	1*	-	-	0	1	**1**	**2**
7	Basic Engineering and Applied Sciences	4	7*	2	1*	1	1	**7**	**9**
8	Physical Education	-	1*	-	-	-	-	-	**1**
	Total number of posts	**17**	**24**	**8**	**4**	**6**	**8**	**31**	**36**

Staff from Agricultural College/Research/Extension and Contractual faculty
SS - Sanctioned Strength
IP - In Position

6.4.3 Technical and Supporting Staff

A separate office is available for the administration and coordination of the M.Tech. (Farm Machinery and Power Engineering) programme. The office is responsible for managing academic, administrative, and student-related activities associated with the programme, including

maintenance of records, communication, scheduling, and coordination of departmental activities. In addition to the dedicated office facilities, the Office and Laboratory Staff of the College extend their support for the smooth and effective functioning of the M.Tech. (Farm Machinery and Power Engineering) programme. The office staff assist in handling administrative and academic documentation, while the laboratory staff support practical classes, research activities, operation and maintenance of laboratory equipment, and other technical requirements related to the programme. Their collective support ensures the efficient conduct of teaching, research, and other academic activities of the department.

The College is sanctioned with 10 permanent positions as technical and supporting staff apart from 26 positions working on Time scale/Outsourcing/Contract basis. The technical and supporting staff of 36 is sufficient to carry out the daily required turnout is more than the number of positions recommended by the Indian Council of Agricultural Research, New Delhi.

Sanctioned and In-Position Office Staff at Dr. NTR CAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Recommended Office Staff by ICAR
1	Administrative Officer	1	1	-	1
2	Superintendent	1	1	-	20 Clerks, steno, Messenger
3	Senior Assistant	2	2*	-	
4	JACT (Junior Asst cum Typist)	2	1+7*	-	
5	Record Assistant	1	1	-	
6	Office Subordinate	2	2	-	
7	Office supporting staff	-	8*	-	-
8	Messenger/Peon	-	10*		8
9.	Watchman	1			
	Total	**10**	**36**	**-**	**29**

** Staff recruited on Time scale/Outsourcing/Contract*

The College has a sanctioned strength of 16 permanent positions as technical support to laboratories whereas a total of 41 personnels are working on Time scale/ Outsourcing/ Contract basis. The Laboratory staff of 41 is sufficient to carry out the daily required turnout, which is

equal to the number of positions recommended by the Indian Council of Agricultural Research, New Delhi.

Table 4. Sanctioned and In-Position Laboratory Staff at Dr. NTR CAE, Bapatla

Sl. No.	Sanctioned Post / Category	Sanctioned Strength	In-Position	Vacant Positions	Laboratory Staff Recommended by ICAR
1	Workshop Supervisor	1	1*	-	27 Workshop Staff, Laboratory Staff, Driver, Technicians
2	Technician (Mech. Gr.I)	3	5*	-	
3	Technician (Mech.Gr.II)	2	2*	-	
4	Draughtsman Grade III	1	1*	-	
5	Carpenter Grade II	1	1*	-	
6	Welder	1	1*	-	
7	Blacksmith	1	1*	-	
8	Turner	1	1*	-	
9	Driver	1	1+3*	-	
10	Workshop Helper	2	2*	-	
11	Luskar	2	2*	-	
12	Watch & Ward Staff	0	8*	-	
12	Laboratory Assistant	-	8*	-	14
13	Computer operator	-	3*	-	
	Total	**16**	**41**	**-**	**41**

** Positions filled with Time scale/Outsourcing/Contractual employees*

6.4.4 Classrooms and Laboratories

The Department of Farm Machinery and Power Engineering is equipped with the requisite instructional infrastructure to effectively administer the M.Tech. program in Farm Machinery and Power Engineering. The Department includes a dedicated classroom specifically for theoretical instruction within the doctoral program. The classroom is spacious, well-lit, and adequately ventilated, providing a comfortable and conducive environment for teaching and learning activities. It is furnished with essential teaching aids and modern instructional facilities, including an LCD projector, a chalkboard, appropriate furniture, and audio-visual support systems, to facilitate effective classroom instruction. Faculty members regularly utilize these facilities to deliver lectures, presentations, seminars, and interactive discussions. The classroom infrastructure supports both traditional and ICT-enabled teaching methods, in accordance with

ICAR recommendations and contemporary pedagogical practices. In addition to classroom facilities, the Department maintains well-established laboratories to support practical instruction, advanced experimentation, and research activities related to Farm Machinery and Power Engineering. These laboratories are regularly employed to conduct practical classes, laboratory exercises, demonstrations, student research work, and testing of agricultural machinery and equipment. The laboratories are equipped with the necessary instruments, equipment, tools, and experimental setups pertinent to the curriculum and research needs of postgraduate and doctoral students. The major laboratories available in the Department are listed below:

1. Farm Power Laboratory
2. Farm Machinery Laboratory
3. Advanced Workshop
4. Soil Dynamics Laboratory
5. Ergonomics Laboratory
6. Field laboratory

The Farm Power Laboratory provides practical training and research opportunities centred on tractors, engines, power transmission systems, and renewable energy applications in agriculture. The Farm Machinery Laboratory is equipped for testing, calibrating, operating, and evaluating the performance of various agricultural machines and implements. The Advanced Workshop offers facilities for fabrication, machining, repair, maintenance, and prototype development activities related to farm machinery and equipment. The Soil Dynamics Laboratory supports research and experimentation concerning soil-tool interaction, tillage mechanics, traction studies, and soil physical properties. The Ergonomics Laboratory is used for studies on human engineering, operator comfort, safety, physiological workload, and ergonomic evaluation of agricultural tools and machinery. These laboratories play a crucial role in enhancing experiential learning, hands-on training, and research competency in students. The Department continuously upgrades its laboratory facilities and equipment to meet the emerging needs of agricultural mechanization research and maintain the standards prescribed by the ICAR and the University. The available infrastructure significantly contributes to improving the quality of teaching, research, innovation, and technology development in the Department of Farm Machinery and Power Engineering.

Table 5. Laboratories available for PG Programme

Sl. No.	Name of the laboratory	Name of the Important Equipment/Instruments
1	Farm Power Laboratory	Cut models - Diesel engine (single cylinder), Tractor differential, Tractor steering, Rear axle assembly, Hydraulic brake, Master cylinder, Ignition system, Self-starter, Distributor, Ignition Coil, Magneto, Synchromesh gear box, Constant mesh gear box, Steering gear box, diaphragm clutch system, multi plates clutch, Four wheel brake, cone clutch, mechanical brake system. LED Panels - Fuel injection system, Lubrication system, Gear box 8+2, Hydraulic system., Tractor cut section without wheel assemblies, etc.
2	Farm Machinery Laboratory	Knapsack sprayer, Hand compression sprayer, Mist blower, Hand operated castor thresher, Groundnut pod stripper, Axial flow thresher, Paddy reaper, Maize Sheller, Power Sprayer, Sub-soiler, Paddy thresher, ULV Sprayer, Post hole digger, 9 tyne cultivator (spring loaded), Boom sprayer, Disc plough, Battery operated cotton picker, Wheel type dibbler, Rotary mulcher, Groundnut digger shaker cum windrower, Mobile shredder, Foot operated sprayer, Rocking sprayer, Bucket scraper, Multi crop thresher (Electric motor operated), Rotavator, Rice transplanter (Riding type), Rice transplanter (walking type), Nursery raising machine, Seed cum fertilizer drill automatic, Reversible M.B. Plough, Rotary plough, Spider weeder, Multi crop ridge planter, Rotary slasher, Laser land leveler, Self-propelled high clearance sprayer, Combine harvester (self-propelled), Combine harvester (tractor mounted), Rigid tyne cultivator, John Deere 4 WD tractor along with hoe and bucket attachments, Mahindra 40 H.P, Escorts tractor, V.S.T Mini tractor, Power weeder, etc.
3	Advanced Workshop	Lathe machines with all accessories and attachments, Spot welding machine (pedestal type), Welding Machine with transformer (600 AMPS air cooled)

		and accessories, Wooden work table (3’ X 8’), Bench grinder ½ HP, Lathe machines, Wood turning lathe, Shaper machine (18”), Air compressor, Iron cutting machine (14”), Hydraulic press, Shearing machine, Hydraulic hack saw machine with 2 hp motor, Hydraulic pipe bending machine, Stand drilling machine ½”, Pipe bending machine (½” to ¼”), Hand drilling machine (6 mm), Universal milling machine, Radial drilling machine 1” capacity, Heavy duty sheet bending machine (4’), CNC Lathe machine, CNC Milling machine, etc.
4	Soil Dynamics Laboratory	Rectangular and circular soil bins, Force analysis instrumentation, S Type load cells (1 ton and 3 ton capacity), Digital force gauge, Digital and analogue tachometers, Cone penetrometer, Strain gauge (Canti-lever type), Universal load cell (750 kg), Digital transducer indicator suitable for AG make universal load cell, Variable frequency drive motor (1.5 kW), Sensors for rectangular soil bin- PH sensor, temperature humidity sensor, wind speed sensor, ultra sonic sensor, hall sensor and rpm sensor; Universal Data Logger system, inputs: 8-10 , free standing panel type, PLC based, Ethernet facility; Touch screen display system- 10 Inch screen display, Ethernet and comport access, data storage mode in pen drive; PLC based software for universal data logger with graphical display development and HMI screen, etc.
5	Ergonomics Laboratory	Treadmill, Ergo cycle, Mini hand-held pulse oximeter, Heart rate sensor, Digital blood pressure monitor, Grip strength dynamometer, Ergonomic height scale stand, Kit for measurement of displacement using LVDT, Stain gauge with cantilever arrangement, Vibration meter, Sound level meter, Digital non-contact type tachometer etc.



Fig. 1. Virtual classroom



Fig. 2. Lathe machines



Fig. 3. CNC Lathe machine



Fig. 4. Tractor cut section



Fig. 5. Tractor systems cut sections



Fig. 6. Different sowing equipment



Fig. 7. Different tillage equipment



Fig. 8. Rectangular Soil bin



Fig. 9. Circular Soil bin



Fig. 10 Ergonomics Laboratory



Fig. 11. Field testing of developed machinery

6.4.5 Conduct of Practical and Hands-on Training

Practical classes for the M.Tech. programme in Farm Machinery and Power Engineering are conducted regularly in the respective laboratories as per the approved class timetable. The Department ensures effective practical exposure to students through well-planned laboratory sessions utilizing the available instruments, equipment, machinery, tools, and experimental facilities. The practical curriculum is designed to complement theoretical instruction and to strengthen the analytical, technical, and research capabilities of the students. The laboratories are adequately equipped to provide hands-on training in various areas of Farm Machinery and Power Engineering, including farm power systems, agricultural machinery operation and testing, soil dynamics, ergonomics, workshop practices, instrumentation, and precision agriculture applications. Faculty members and technical staff supervise the practical sessions to ensure proper utilization of laboratory resources and to facilitate experiential learning among students.

Special emphasis is given to developing practical competence, problem-solving ability, and research aptitude through laboratory experiments, demonstrations, field observations, equipment handling, calibration exercises, and data analysis. Students are encouraged to actively participate in experimental work, machinery testing, prototype evaluation, and research-oriented activities relevant to current challenges in agricultural mechanization and farm engineering. As part of experiential learning and exposure to advanced technologies, students are deputed to reputed ICAR research institutes, Central Government organizations, and industries for a period of one month to undergo hands-on training during the Second Year First Semester. These training programmes provide valuable exposure to modern research facilities, advanced instrumentation, industrial practices, and emerging technologies in agricultural engineering. The major institutions and organizations where students are deputed include:

1. ICAR – Central Research Institute for Dryland Agriculture (CRIDA), Hyderabad
2. ICAR – Central Institute of Agricultural Engineering (CIAE), Bhopal
3. ICAR – CIAE Regional Centre, Coimbatore
4. Central Institute of Tool Design (CITD), Hyderabad
5. Southern Region Farm Machinery Training and Testing Institute (SRFMTTI), Garladinne
6. Relevant agricultural engineering industries and manufacturing units

These training programmes enable students to gain practical knowledge and exposure in areas such as farm machinery design and testing, precision agriculture, conservation agriculture, instrumentation, manufacturing processes, automation, renewable energy systems, and agricultural mechanization technologies. Interaction with scientists, engineers, industrial experts, and technical personnel during the training period enhances the students' professional skills, confidence, and understanding of real-time field and industrial problems. The Department considers industrial and institutional exposure as an integral component of the academic programme to bridge the gap between classroom teaching and field application. Such initiatives significantly contribute towards skill development, research orientation, industry interaction, and employability of the students in accordance with the objectives and quality standards prescribed by ICAR and the University. The details of M.Tech. students who took practical training from various institutes during last five years is given below.

Table 6. Details of M.Tech. students undergone practical training

S. No.	Name of M.Tech. Student	ID No.	Name of Training Institute	Duration
2020-21				
1	Maheswari. P	BEM/2020-01	ICAR – Central Institute of Agricultural Engineering (CIAE), Regional Centre, Coimbatore, Tamil Nadu	01.10.2021 to 31.10.21 (1 month)
2021-22				
1	Manoj Kumara	BEM/2021-01	SRFMT&TI, Tractor Nagar, Garladinne, Anathapuramu (Dist.), Andhra Pradesh	10.11.2022 to 10.12.2022 (1 month)
2	Sindhuja. G	BEM/2021-02		
3	Venkatesh. K	BEM/2021-03		
2022-23				
1	Koteswara Rao. K	BEM/2022-01	Central Institute of Tool Design (CITD), MSME – Tool Room, Hyderabad	06.12.2023 to 05.12.2023 (1 month)
2	Madhu Malathi. K	BEM/2022-02		
2023-24				
1	Ramavath Eliya	BEM/2023-01	ICAR- Central Institute of Agricultural Engineering (CIAE), Bhopal	01.03.2026 to 31.03.2026 (1 month)
2	Prahlad. N	BEM/2023-02		
2025-26				
1	Shaik Samreen	BEM/2025-01	ICAR- Central Institute of Agricultural Engineering (CIAE), Bhopal	15.09.2026 to 14.10.2026 (1 month)
2	Swetha. V.	BEM/2025-02		

6.4.6 Supervision of Students in M.Tech. programme

Each M.Tech. student shall be assigned a faculty member from the Department of Farm Machinery and Power Engineering by the Head of the Department, in consultation with the University Head of the Department, to supervise the student's course work and research activities. The assigned faculty member shall serve as the Chairperson of the Student Advisory Committee. In addition, two Advisory Members shall be appointed by the Dean of Post Graduate Studies based on the recommendation of the Chairperson through the Head of the Department. Of these, one member shall be from the Department of Farm Machinery and Power Engineering, while the other shall be from a minor department under Agricultural Engineering or Agriculture, depending on the student's academic and research requirements. The Student Advisory Committee shall guide the student in the selection of courses in the major and minor fields, identification of the thesis research problem, and all other academic matters. Students shall complete course work during the first two semesters and undertake research work during the subsequent two semesters. The progress of each student's research work shall be reviewed monthly by the Student Advisory Committee along with the Head of the Department to provide necessary guidance and suggestions. During the third semester, each student shall appear for a comprehensive viva-voce examination conducted by an external Professor in Farm Machinery and Power Engineering from another State Agricultural University or ICAR Institute nominated by the Dean of Post Graduate Studies. The examination shall assess the student's proficiency in course work and thesis research. The Student Advisory Committee shall also supervise the execution of research work, experimentation, data analysis, and thesis preparation.

Upon completion of the thesis, a pre-submission seminar shall be conducted by the Head of the Department in the presence of the Student Advisory Committee and faculty members of the department to obtain suggestions and corrections, if any. After incorporating the recommended corrections, the thesis shall be forwarded to the Dean of Post Graduate Studies through the Head of the Department. The Dean of Post Graduate Studies shall send the thesis to an external Professor from another State Agricultural University or ICAR Institute for evaluation. Based on the examiner's recommendations, the thesis may be accepted with minor or major corrections or rejected for the award of the M.Tech. degree. After receiving the evaluation report and incorporating all suggested corrections, the student shall present a final submission

seminar before the Student Advisory Committee and faculty members to verify compliance with the examiner's recommendations. Finally, the corrected thesis shall be submitted to the Head of the Department and the Dean of Post Graduate Studies for the award of the M.Tech. degree. The details of M.Tech. students who were supervised by the faculty under Dept. of Farm Machinery and Power Engineering during last five years is given below.

Table 7. List of M. Tech. students' projects and supervisors

S. No.	Name of M.Tech. Student	ID No.	Chairman of the Advisory Committee	Title of Thesis Research Work
2020-21				
1	Ms. Nagavarapu Anusha	BEM 19-02	Dr. K.V.S. Rami Reddy, Professor	Development of Sensor based paddy planter for automatic seeding.
2	Mr. Basani Nagaraj	BEM 19-03	Dr. A. Ashok Kumar Assistant Professor	Development of remote control operated sensor based site specific chemical applicator.
3	Mr. Kure Santosh Kumar	BEM 19-04	Dr. C. Ramana Principal Scientist & Univ. Head (FMPE)	Development of high clearance self-Profelled Chilli Harvester.
4	Mr. Batta Suresh	BEM 19-05	Dr. B. Hari Babu Professor & Head	Development and evaluation of multipurpose Equipment for row crops.
2021-22				
1	Maheswari. P	BEM/2020-01	Dr. A. Ashok Kumar, Assistant Professor	Study the Feasibility of Zero-Till Happy Seeder Under Different Soil Conditions of Andhra Pradesh
2022-23				
1	Manoj Kumara	BEM/2021-01	Dr. A. Srinivasa Rao, Assistant Professor	Design and Development of a Microcontroller –Based Variable Rate Sprayer for Field Crops
2	Sindhuja. G	BEM/2021-02	Dr. A. Ashok Kumar, Assistant Professor	Development of Micro-Controller Based Roto Planter For Direct Sowing of Paddy Suitable for Mini

				Tractor
3	Venkatesh. K	BEM/2021-03	Dr. B. Raj Kiran, Assistant Professor	Development and Performance Evaluation of Tractor Operated Two Row Semi-Automatic Vegetable Transplanter
2023-24				
1	Koteswara Rao. K	BEM/2022-01	Dr. B. Raj Kiran, Assistant Professor	A Real Time Vision Based Inter Row Weed Detection and Herbicide Spraying System.
2	Madhu Malathi. K	BEM/2022-02	Dr. A. Srinivasa Rao, Assistant Professor	Development of Remote –Control Operated Self-Propelled Groundnut Planter
2024-25				
1	Ramavath Eliya	BEM/2023-01	Dr. D. Vijay Kumar, Assistant Professor	Development and Performance Evaluation of Automatic Inter and Intrarow Weeder for Wide Spaced Crops
2	Prahlad. N	BEM/2023-02	Dr. G. Veera Prasad, Scientist & Principal Investigator, AICRP on FIM	Design and Development of Robotic Pneumatic Planter for Cotton
2025-26				
1	Shaik Samreen	BEM/2025-01	Dr. A. Srinivasa Rao, Assistant Professor	Design and Development of Mechatronic Metering Based Urea Super Granular (USG) Applicator
2	Swetha. V.	BEM/2025-02	Dr. D. Vijay Kumar, Assistant Professor	Development of Finger Weeder For Intra Row Weeding with Crop Row Detection System

6.4.7 Feedback of Stakeholders (Students, Parents, Industries, Employers, Farmers, etc.)

The M.Tech. program in Farm Machinery and Power Engineering at Dr. N.T.R. College of Agricultural Engineering, Bapatla, has garnered commendable feedback from students regarding its academic environment, teaching standards, research guidance, and infrastructural facilities. The Department is dedicated to maintaining academic excellence through quality teaching, effective mentoring, practical exposure, and research-oriented learning, which has been

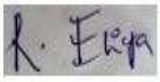
positively acknowledged by students and other stakeholders. The academic program is implemented in accordance with the ICAR guidelines and university regulations, focusing on developing technical competence, research aptitude, innovation, and professional skills among scholars. Students have expressed appreciation for the supportive learning atmosphere, accessibility of faculty members, well-equipped laboratories, field exposure, seminar activities, and opportunities to interact with research institutes and industries. The regular conduct of seminars, laboratory practical, institutional training, and research reviews has significantly contributed to strengthening their academic and research capabilities. Feedback from alumni indicates that the program has equipped them with advanced knowledge and practical expertise in various domains of Farm Machinery and Power Engineering, enabling them to pursue successful careers in teaching, research, government organizations, industries, and allied sectors. Alumni have also expressed satisfaction with the quality of faculty guidance, the availability of academic resources, and exposure to contemporary technologies and research methodologies.

Progressive farmers and stakeholders involved in extension and technology dissemination activities have commended the Department's efforts in developing and transferring appropriate farm mechanization technologies that are beneficial to the farming community. The program's practical orientation and scholars' research have addressed field-level agricultural engineering challenges and promoted sustainable agricultural mechanization. Training organizations, research institutions, and recruiting agencies have also provided positive feedback regarding the professional competence, technical knowledge, communication skills, and practical exposure of students graduating from the program. Employers and collaborating organizations have recognized graduates' ability to contribute effectively to research, teaching, extension, consultancy, and industrial applications related to agricultural engineering. The feedback received from various stakeholders is regularly reviewed by the Department and utilized for the continuous improvement of the academic program, curriculum delivery, research activities and student support services. Such stakeholder interactions and feedback mechanisms play a crucial role in enhancing the overall quality and effectiveness of the M.Tech. program in Farm Machinery and Power Engineering. A selection of representative feedback from alumni, progressive farmers, training organizations, and recruiting organizations is presented below:

Feedback from M.Tech. (Farm Machinery and Power Engineering) Alumni Students

I am Ramavath Eliya, I completed my M.Tech. in Farm Machinery and Power Engineering from Dr. NTR College of Agricultural Engineering, Bapatla during 2023-2025. Currently, I am pursuing my Ph.D. at the Central Institute of Agricultural Engineering (CIAE), Bhopal. My journey at Dr. NTR College of Agricultural Engineering, Bapatla had been academically enriching and professionally transformative, greatly shaping my research interests and career aspirations. The department provided an excellent environment for learning, innovation and practical exposure, which laid a strong foundation for my higher studies and research pursuits. The faculty members, with their vast knowledge, constant encouragement and valuable mentorship, always motivated students to develop technical competence, analytical thinking and professional confidence. Their guidance significantly enhanced my research aptitude, presentation skills and practical understanding of the subject. In addition, the well-equipped laboratories, workshop facilities and research-oriented atmosphere offered meaningful opportunities for experiential learning and technical development. I sincerely express my gratitude to all the teaching and non-teaching staff members for their continuous support and encouragement throughout my academic journey. I will always cherish with great pride my association with this esteemed institution.

Yours sincerely



Feedback received from Mr. Ramavath Eliya, SRF -2nd rank (ST category) & Overall AIR-9 currently perusing Ph.D (Farm Machinery and Power) at CIAE, Bhopal during academic year 2025-26

I am Maheshwari Patil from Karnataka and I completed my M.Tech. in the year 2022 from the Department of Farm Machinery and Power Engineering at Dr. NTR College of Agricultural Engineering, Bapatla under the guidance of Dr. A. Ashok Kumar. Assistant Professor. My postgraduate journey at this esteemed institution was highly valuable and enriching due to the constant guidance and encouragement provided by the faculty and staff members. Their systematic teaching methods, practical exposure, and technical guidance greatly improved my knowledge, research aptitude, communication skills, and self-confidence during my coursework and thesis work. The Department of Farm Machinery and Power Engineering offers an excellent academic atmosphere with advanced laboratories, workshop facilities, and research infrastructure, which helped me successfully carry out experiments and project activities. I sincerely thank all the teaching and non-teaching staff members for their encouragement, cooperation, and valuable support throughout my M.Tech. program.



Your's Sincerely,
Maheshwari
PhD Scholar
CAE, UAS, Raichur

Feedback received from Miss. Maheswari Patil, Alumni

I am Sindhuja Gone, and I completed my M.Tech. during 2023 from the Dept. of Farm Machinery and Power Engineering at Dr. NTR College of Agricultural Engineering, Bapatla. My two-year postgraduate journey at this institution was smooth, enriching and memorable due to the continuous support and encouragement extended by the faculty and staff members. The faculty members are highly knowledgeable, experienced and always willing to guide students in both academic and research activities. Their effective teaching methodology and practical approach significantly enhanced my technical knowledge, research capabilities, presentation skills and overall confidence throughout my coursework and thesis work. The Department of Farm Machinery and Power Engineering provided an excellent academic environment with well-equipped laboratories, workshop facilities and research infrastructure, which enabled the effective conduct of experiments and my project work. I sincerely express my gratitude to all the faculty members and staff for their constant support, encouragement, and motivation throughout my M.Tech. journey.

G. Sindhuja
Ph.D Scholar (FMPE)
Dr-NTR CAE, Bapatla

Feedback received from Miss. Sindhuja Gone, Alumni

Feedback of MTech. (Farm Machinery and Power Engineering) Passed Out Student

I, KODURU KOTESWARA RAO, completed my MTech. in the Department of Farm Machinery and Power Engineering during the academic years 2022–2024 at Acharya N. G. Ranga Agricultural University, Dr. N.T.R. College of Agricultural Engineering, Bapatla. I would like to express my sincere gratitude and positive feedback regarding my alma mater and the department.

The department provided me with exceptional academic guidance, research support, and continuous motivation throughout my master's program. The faculty members always trusted my abilities and encouraged me to explore advanced research areas such as machine vision-based systems, precision agriculture, and intelligent farm machinery. They also supported my collaborations and interactions with researchers from other universities and institutions, which significantly enhanced my technical knowledge and research exposure.

One of the greatest strengths of the department was the strong encouragement provided for innovation and student development. I was given opportunities to serve as a student-teacher member and organizing secretary for the National PG Students Conference, which improved my leadership, communication, and professional skills. The department also provided sufficient research funding, a good stipend throughout my program, and financial support for conducting experimental and field research activities.

The institution has excellent infrastructure facilities, including advanced workshops equipped with modern manufacturing tools and highly skilled technical staff with immense field-level experience. The advanced computer laboratories with CAD and software development facilities greatly supported my master's research work. In addition, the college provided a well-equipped library, PG cabins with Wi-Fi facilities, and comfortable hostel accommodation with spacious rooms and a good academic environment.

Most importantly, the faculty members strongly motivated and guided me in preparing for the ICAR-SRF examination. With their continuous support, encouragement, and blessings, I secured All India Rank 3 in my category and achieved admission into ICAR-Indian Agricultural Research Institute, one of India's premier agricultural institutes. I will always remain thankful to the department, faculty members, technical staff, and institution for shaping my academic and research career.

Thanking you sir,

Yours sincerely,



Er. KODURU KOTESWARA RAO

Feedback received from Mr. K. Koteswara Rao, SRF- 48th and 3rd rank (SC category) currently pursing Ph.D (2024-25) at IARI , New Delhi

I, KODURI MADHU MALATHI, completed my M.Tech. in the Department of Farm Machinery and Power Engineering during the academic years 2022–2024 at Acharya N. G. Ranga Agricultural University, Dr. N.T.R. College of Agricultural Engineering, Bapatla. I would like to express my sincere feedback and gratitude towards the college for providing an excellent academic and professional environment throughout my postgraduate studies.

The college has highly qualified and dedicated faculty members, each possessing strong expertise in their respective subjects. Their teaching methodologies helped students gain both theoretical and practical knowledge from fundamental concepts to advanced applications in agricultural engineering. The faculty members continuously motivated and guided students in academic activities, research work, and competitive examination preparation, especially for ICAR-SRF and other national-level examinations.

The institution provides excellent academic facilities, including a well-equipped library and advanced postgraduate computer laboratories that supported research, software applications, data analysis, and technical learning. Regular seminars, technical sessions, research review meetings, and expert guest lectures organized by the college greatly contributed to improving research knowledge, communication skills, and career development. The college also encouraged students to participate in national-level trainings and workshops at reputed institutes, which enhanced practical exposure and professional confidence.

Apart from academics, the college strongly focused on extracurricular and co-curricular activities for the overall personality development of students. Students were encouraged to participate in cultural activities, sports, leadership programs, and technical events, which helped in building teamwork and communication skills.

The college also provided financial support in the form of stipends and research funding for conducting experimental and project work. In addition, the hostel facilities were very good, with Wi-Fi connectivity, television, healthy food, spacious rooms, playground facilities, and a student-friendly environment that supported comfortable living and academic concentration.

I am thankful to the college administration, faculty members, technical staff, and non-teaching staff for their continuous encouragement, support, and guidance during my postgraduate studies.

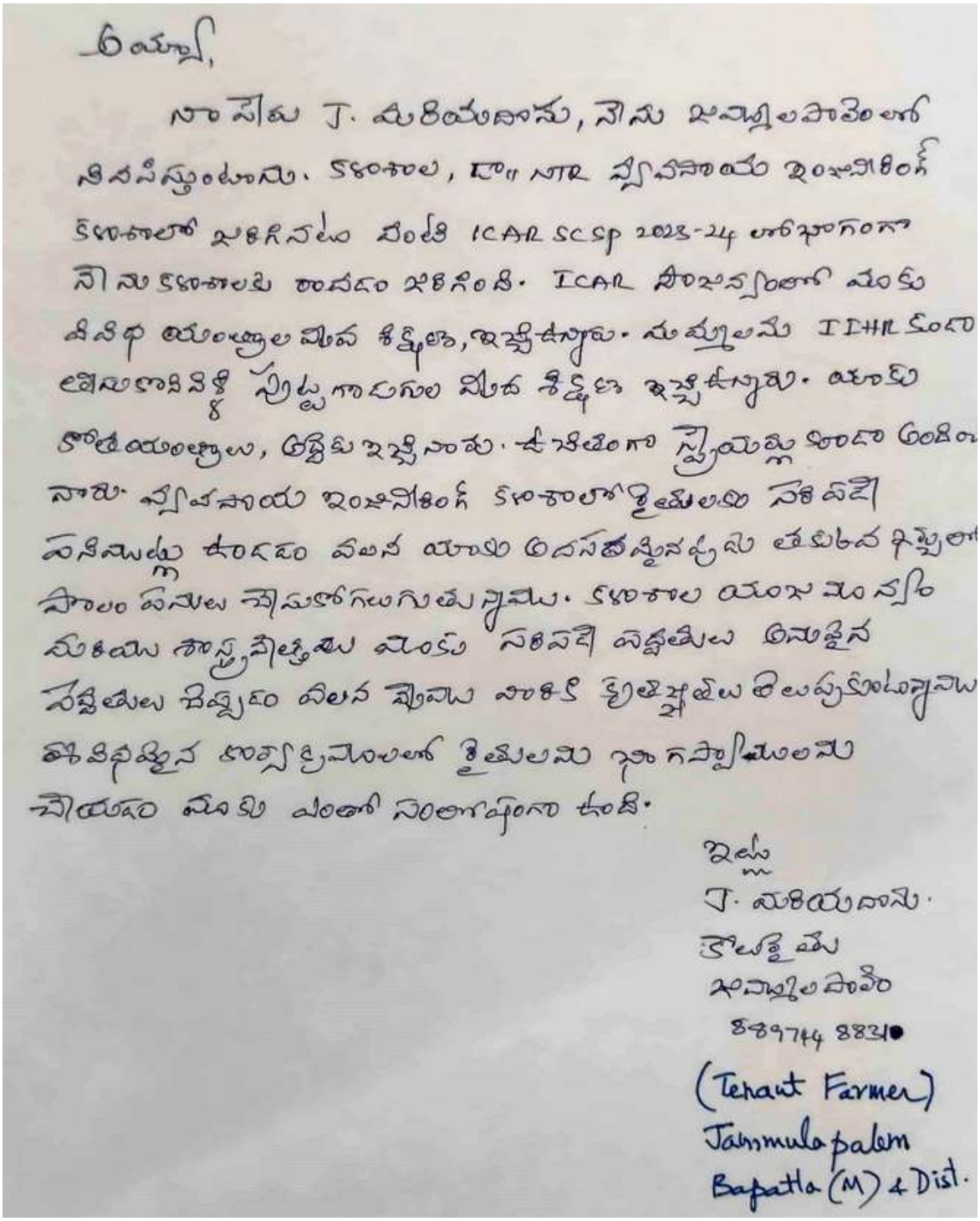
Yours Sincerely,

K. Madhu Malathi

Er. KODURI MADHU MALATHI

Feedback received from Miss. K. Madhu Malathi, SRF – 19th Rank & 12th rank (OBC) currently pursuing Ph.D (2024-25) at IARI , New Delhi

Feedback from Progressive Farmers

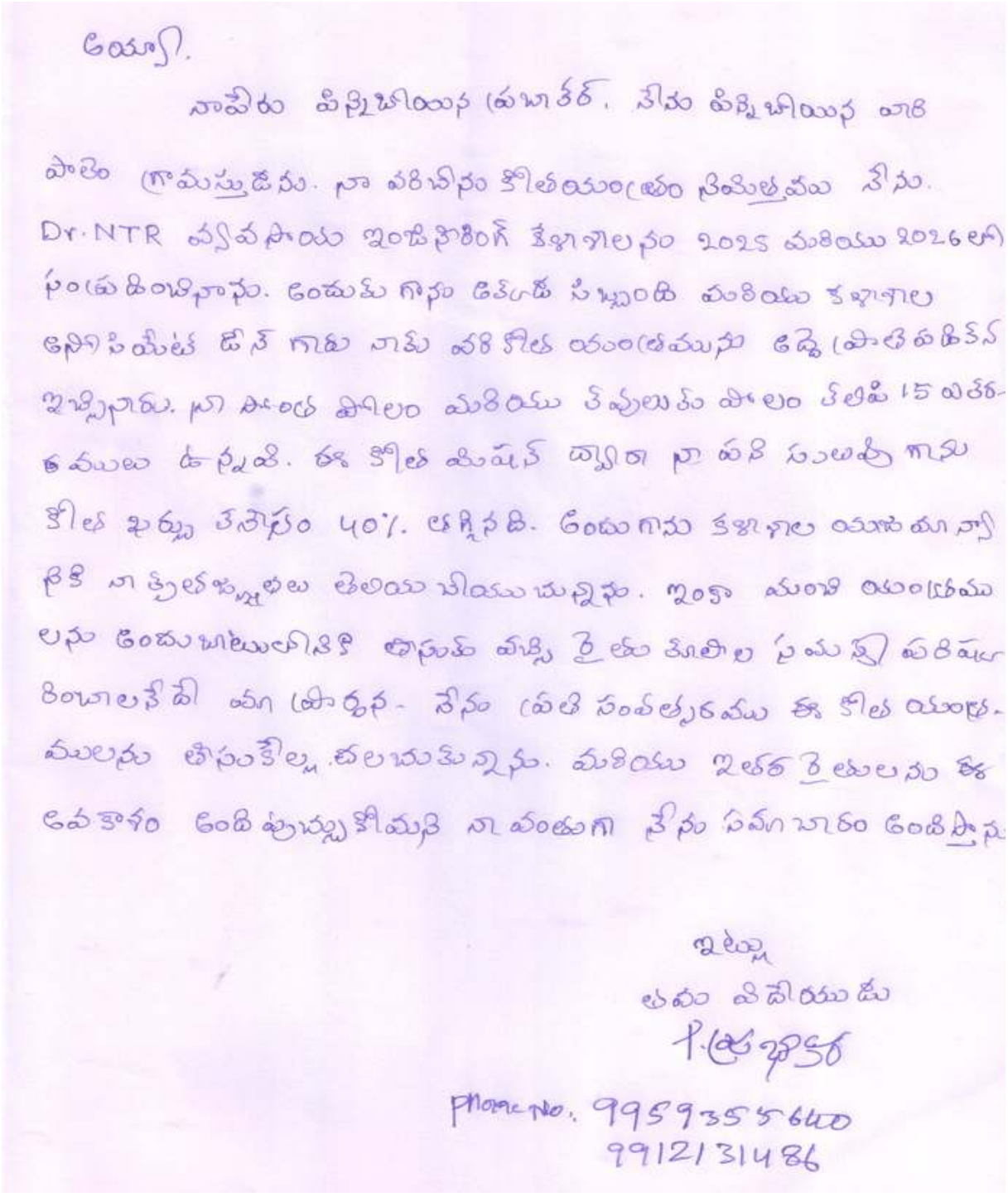


అయ్యా,

నా పేరు J. మరియదాసు, నేను జమ్ములపాలెంలో నివసిస్తుంటాను. కళింశాల, డా॥ NTR వ్యవసాయ ఇంజనీరింగ్ కళాశాలలో జరిగినటు వంటి ICAR SCSP 2023-24 లో భాగంగా నేను కళింశాలకు రావడం జరిగింది. ICAR వారివల్లంతో మంకు వివిధ యంత్రాల మీద శిక్షణలు, ఇచ్చేఉన్నారు. ముఖ్యంగా IIHR కంటా లాబొరేటరీ స్థాపించి రైతుల మీద శిక్షణలు ఇచ్చేఉన్నారు. అలాగే కోతియంత్రాలు, అద్దెకు ఇచ్చినారు. ఈవిధంగా వ్యవసాయంలో చాలా అభివృద్ధి నారు. వ్యవసాయ ఇంజనీరింగ్ కళాశాలలో శిక్షణలు సరిపడే పనిముట్లు ఉండడం వలన అలాగే అందజేస్తున్నట్లు తక్కువ ధరలో పాలం పనులు చేసుకోగలుగుతున్నాము. కళాశాల యందు మంచి శాస్త్రవేత్తలు మంచు సరిపడే పద్ధతులు అనుసరిస్తూ పద్ధతులు చెప్పడం వలన మేము వాళ్ళకి క్రొత్తవిషయాలు తెలుపుకొంటున్నాము వివిధమైన రూపక్రమంలో శిక్షణలను ప్రాంగణాలను చేయడం మాకు ఎంతో సంతోషంగా ఉంది.

ఇట్లు
J. మరియదాసు.
కౌలురైతు
జమ్ములపాలెం
8897448834
(Tenant Farmer)
Jammulapalem
Bapatla (M) & Dist.

Feedback received from Mr. J. Mariya Dasu, Progressive Farmer



అయ్యా,

నా పేరు పి.ప్రభాకర్. నేను పి.ప్ర.భాయిన వారి పాలెం గ్రామస్తుడను. నా వరిచిన్నం కోతయంత్రం సౌమిత్రవు నేను. Dr. NTR వ్యవసాయ ఇంజినీరింగ్ కేళాశాలనం 2025 మరియు 2026లో పంపు చేసాము. దందుకు గాను డిపార్టు సిబ్బంది మరియు కళాశాల ప్రొఫెసర్ డీన్ గారు నాకు వరి కోత యంత్రమును డ్వాక్రా ద్వారా ఇచ్చినారు. నా సొంత పొలం మరియు కౌలుకు పొలం కలిపి 15 ఎకరాలు ఉన్నవి. ఈ కోత మిషన్ ద్వారా నా వరి సంబంధించి కోత ఖర్చు దాదాపు 40%. తగ్గినది. దందుగాను కళాశాల యాజమాన్యానికి నా కృతజ్ఞతలు తెలియ చేయుచున్నాను. ఇంకా మంచి యంత్రములను దందుబాటులోనికి తెచ్చుకు వచ్చిన రైతు కొలతల ప్రకారం పరిశీలించాలనేది ఎంత ప్రార్థన. నేను ప్రతి సంవత్సరము ఈ కోత యంత్రములను తెప్పించుకొన్నాను. మరియు ఇతర రైతులను ఈ అవకాశం దండిపుచ్చుకోమని నా వంతుగా నేను సహకారం దండిస్తాను.

ఇట్లు
మీ విధేయుడు
P. ప్రభాకర్

Phone No. 9959355440
9912131486

Feedback received from Mr. Prabhakar, Progressive Farmer

Feedback from Recruiting Organization

ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

DEPARTMENT OF AGRICULTURAL ENGINEERING

CONDUCT CERTIFICATE

This is to certify that **Dr. M. Vinayak**, Employee ID: **6089**, is working as an **Assistant Professor** in the Department of Agricultural Engineering, Aditya University since **08-07-2024**. During his tenure in this institution, his conduct and character have been found to be **good**. This certificate is issued based on office records and for the purpose as required by him.

Date: 14-05-2026
Place: Surampalem

(Dr. B. KRISHNA KANTH)

Head of the Department

Department of Agricultural Engineering

Head of the Department
Dept. of Agricultural Engineering
ADITYA UNIVERSITY

Feedback received from recruiting institute Aditya University, Kakinada

ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

DEPARTMENT OF AGRICULTURAL ENGINEERING

CONDUCT CERTIFICATE

This is to certify that **Dr. S. Rahaman**, Employee ID: **5656**, is working as an **Assistant Professor** in the Department of Agricultural Engineering, Aditya University since **03-07-2023**. During his tenure in this institution, his conduct and character have been found to be **good**. This certificate is issued based on office records and for the purpose as required by him.

Date: 14-05-2026
Place: Surampalem

(Dr. B. KRISHNA KANTH)

Head of the Department

Department of Agricultural Engineering

Head of the Department
Dept. of Agricultural Engineering
ADITYA UNIVERSITY

Feedback received from recruiting institute Aditya University, Kakinada

Feedback from Parents

To
The Dean sir,

I the parent of Konduri Madhu Malathi, who have studied the M. Tech in the Department of Farm Machinery and Power Engineering, Bapatla during the academic year 2022-24. It's our pleasure to thank the institute for your consistent motivation in timely completion of her PG work. I thank all the faculty members for their support during the pursuance of her research work. The faculty are cooperative in clearing the concepts and it had helped our daughter to secure All India Rank -19. This fetched her to select the Ph.D at IARI Pusa Campus at New Delhi. The support from the faculty and administration was highly admirable during her stay at Bapatla. The quality of education, food and premises atmosphere, security for girls' students is highly applauded. Once again, I thank all the staff, administration for your support and knowledge imparted to our kid in achieving her dream.

K. Rajeswar.

Konduri Rajeswar
Father of Konduri Madhu Malathi
Julapally Village, Peddapalli (Dist)
Telangana -505525
Phno. 9948255447

Feedback received from father of Miss. K. Madhu Malathi, Alumni

To
The Associate Dean
Dr. NTR College of Agricultural Engineering
Bapatla

I, Subramani, parent of Manoj Kumara, currently perusing Ph.D student (2024-25) of Dr. NTR College of Agricultural Engineering, Bapatla. He also perused his M.Tech in Farm Machinery and Power Engineering from the same college during 2021-23. I take immense pleasure in expressing my heartfelt gratitude to the Department of Farm Machinery and Power Engineering at Dr. NTR College of Agricultural Engineering, where my son pursued her M.Tech during the academic years 2022–2024. The work during his M. Tech program has motivated to take admission in Ph.D program.

We are deeply thankful to the institute for its continuous encouragement, academic guidance, and unwavering support during his Ph.D studies as well. The dedication and commitment shown by the faculty members in mentoring students and strengthening their conceptual understanding played a vital role in shaping her academic excellence. Their timely guidance and supportive approach greatly contributing to prepare for the ARS examinations, ICAR NET. We sincerely appreciate the cordial environment maintained by the faculty and administration during her stay at Bapatla. The institution not only provided quality education and research exposure but also ensured a safe, disciplined, and student-friendly atmosphere. The hostel facilities, food quality, campus ambience, are highly commendable.

It is truly gratifying to witness the efforts of the institution in nurturing students academically and personally. We once again extend our sincere thanks to all the teaching faculty, staff members, and administration for their valuable support.

V. Subrami

Subramani V
Taaduru, Shidalghatta
Chikkaballapura,
Karnataka – 562102
Ph.no. 9632971798

Feedback received from father of Mr. Manjoj Kumara, Alumni

Feedback from Training Institute

இ.வே.ஆ.கு.-மத்திய வேளாண் பொறியியல் நிறுவனம்
மண்டல நிலையம், கோயம்புத்தூர் - 641 007 (தமிழ்நாடு)
भा. कृ. अनु. प. – केन्द्रीय कृषि अभियांत्रिकी संस्थान
क्षेत्रीय केंद्र कोयम्बटूर – 641007 (तमिलनाडु)
ICAR - CENTRAL INSTITUTE OF AGRICULTURAL ENGINEERING
REGIONAL STATION, COIMBATORE – 641007 (TAMILNADU)
Telephone Nos. 0422-2472623 E-mail: rsciaecbe@gmail.com

Dr. Ravindra Naik
Principal Scientist & Head(i/c)

No.CIAE-RS/9-9(E)/2026-27
Dated 18.05.2026

TO WHOMSEVER IT MAY CONCERN

This is to certify that the Regional Station, ICAR–Central Institute of Agricultural Engineering, Coimbatore, has been providing In-Plant Training to the undergraduate, M. Tech., and Ph.D. students of Dr. NTR College of Agricultural Engineering, Bapatla. It is observed with appreciation that the students deputed from the aforementioned institution have demonstrated commendable sincerity and punctuality throughout the duration of their training at this station. They were actively engaged in the ongoing research projects of the institute and have contributed meaningfully to the tasks entrusted to them. The students have also participated earnestly in various institutional activities and have shown a high degree of dedication and commitment towards the successful completion of their assigned responsibilities. Their conduct and academic engagement during the training period have been found to be highly satisfactory.

Dr. Ravindra Naik
Principal Scientist & Head, I/c
தலைவர் / प्रभारी / HEAD
மத்திய வேளாண் பொறியியல் நிறுவனம்
மண்டல நிலையம், கோயம்புத்தூர் 641 007.
भा.कृ.अनु.परि. - केन्द्रीय कृषि अभियांत्रिकी संस्थान
क्षेत्रीय स्टेशन, कोयंबटूर - 641 007
ICAR-CENTRAL INSTITUTE OF AGRICULTURAL ENGINEERING
Regional Station, Coimbatore - 641 007.

Feedback from internship ICAR-CIAE, Regional Center, Coimbatore

JOHN DEERE

M/s. VVC FARM EQUIPMENTS PVT. LTD.,
Phase-IV, Plot No.1, Agathavarappadu,
Indira Auto Nagar, GUNTUR-522 001.
Cell : 7729992036
E-mail : vvcfe.guntur@gmail.com
GSTIN : 37AAECV7159A1ZA, PAN No : AAECV7159A

TO WHOMSOEVER IT MAY CONCERN

We, VVC FARM Equipment PVT LTD, John deere Dealership as started giving training programmes on service, inspection of farm machinery and tractors. The undergraduate and PG students of Farm Machinery and Power Engineering,of Dr. NTR College of Agricultural Engineering, Bapatla has got involved in the activates of commissioning the hands-onpractise on repairs of tractor and machinery. The students are disciplined, punctual in doing their work. During their stay, they did not bring any remarks on the institution.

Thanks & Regards,

VVC Farm Equipments Pvt Ltd.

Feedback from tractor industry intenship organization

6.4.8 Student Intake and Attrition in the M.Tech. Programme in Farm Machinery and Power Engineering for Last Five Years

Admissions to the M.Tech. program in Farm Machinery and Power Engineering are conducted through a transparent, merit-based selection process. Since the 2018–19 academic year, admissions have been facilitated via the Junior Research Fellowship (JRF) entrance examination, which is organized annually by the Indian Council of Agricultural Research (ICAR), while strictly adhering to the reservation policy of the Government of Andhra Pradesh. Before the implementation of the ICAR centralized admission system, Acharya N.G. Ranga Agricultural University (ANGRAU) conducted separate entrance examinations for each specialization based on the prescribed B.Tech. (Agril. Engg.) syllabus. The admission procedure is designed to ensure the selection of academically competent and research-oriented candidates with strong foundational knowledge of agricultural engineering and allied disciplines. The program attracts students from various regions of the country through the ICAR admission system, thereby enhancing academic diversity and promoting an interdisciplinary research culture within the department.

In addition to regular admissions, 15% of the total intake is allocated under the ICAR quota, wherein students are admitted with ICAR fellowship support. This fellowship enables scholars to pursue high-quality research in priority areas such as farm machinery, renewable energy applications, precision farming, mechanization, automation, and sustainable agricultural production systems. The Department has consistently maintained a satisfactory level of student enrolment and research output over the years. During the last five academic years, 16 students successfully completed and were awarded M.Tech. degrees in Farm Machinery and Power Engineering. The research conducted by these scholars has contributed significantly to the development of location-specific technologies, farm mechanization practices, energy-efficient systems, and sustainable agricultural engineering solutions relevant to the farming communities.

Table. 8. Student Intake and Attrition for the College in M.Tech. (Farm Machinery and Power Engineering) Programme

Name of Degree Programme	Actual student admitted in last five years						Attrition (%)					
	Y 1	Y2	Y3	Y4	Y5	Y6 *	Y1	Y2	Y3	Y4	Y5	Y6 *
M.Tech. – Farm Machinery & Power Engineering	1	3	2	2	0	2	0	0	0	0	0	0

** Current year 2025-26*

6.4.9 ICT Application in Curricula Delivery

The Department of Farm Machinery and Power Engineering is equipped with state-of-the-art teaching and learning facilities that enhance the delivery of academic programs and foster an interactive-learning environment. The Department features smart classrooms with interactive boards, multimedia projectors, audio-visual systems, Internet connectivity, and other ICT-enabled teaching aids for classroom instruction, seminar presentations, and review meetings related to students' research and project work. Faculty members regularly utilize these resources to deliver course lectures using PowerPoint presentations, multimedia content, video demonstrations, and other digital learning tools. The integration of modern teaching tools has significantly improved classroom instruction by enhancing conceptual understanding, student engagement, and the interaction between teachers and students. The Department also employs learner-centric teaching methodologies that encourage active participation, analytical thinking, and problem-solving among students. Students are encouraged to deliver seminar, project, and research progress presentations using Power Point and other digital platforms. This practice helps improve technical communication skills, presentation abilities, confidence, and overall professional competence. Regular seminars, technical discussions, and presentations are organized as part of the curriculum to expose students to recent advances and emerging trends in agricultural engineering. The Department continuously strives to strengthen its academic infrastructure in line with ICAR recommendations and contemporary educational requirements. The available teaching facilities provide a conducive academic atmosphere for effective teaching, learning, research, and knowledge dissemination. These initiatives collectively

contribute to maintaining quality education and achieving academic excellence in Farm Machinery and Power Engineering.

Credentials of the Faculty

Faculty at Dr NTR College of Agricultural Engineering, Bapatla, are well qualified. Apart from Associate Dean, out of 15 faculties, thirteen have Ph.D. qualifications, the rest two have a minimum qualification of M.Tech. in Agricultural Engineering.

Table 9. Educational Qualifications and Experience of the Faculty in the Department

S. No.	Faculty	Designation	Qualification	Experience
Farm Machinery and power Engineering				
1	Dr. K.V.S. Rami Reddy	Professor	Ph.D.	**24**
2	Dr. D. Vijay Kumar	Assistant Professor	Ph.D.	**13**
3	Dr. A. Srinivasa Rao	Assistant Professor	Ph.D.	**13**
4	Dr. B. Raj Kiran	Assistant Professor	Ph.D.	**13**
5	Dr. A. Ashok Kumar	Assistant Professor	Ph.D.	**13**
6	Er. N. Kishore	Assistant Professor	M. Tech.	**9**
7	Dr. G. Veera Prasad	Scientist	Ph.D.	**13**
Processing and Engineering				
8	Dr. S. Vishnu Vardhan	Professor	Ph.D.	**20**
Applied Sciences				
9	Dr. K. Kiran Prakash	Professor	Ph.D.	**18**
10	Dr. M. Lavanya Kumari	Professor	Ph.D.	**17**
11	Dr. T. Prashanth Kumar	Professor	Ph.D.	**25**
12	Dr. Lal Ahmed Mohammad	Professor	Ph.D.	**20**
13	Dr. M. Latha	Professor	Ph.D.	**21**
14	Dr. T. Srinivasa Rao	Associate Professor	Ph.D.	**15**
15	Mr. G. Sesha Reddy	Teaching Associate	M.Sc.	**4**

Table 10. Training programmes/Short course/Workshops/Conferences Organized

Year	Title of the Training programme/ Short course/ Workshop/Conference	Period		Venue	Organized by	Source of Funding	No. of Participants
		From	To				
2021	Brainstorming session on “Mechanization and Automation for	29.12. 2021	29.12. 2021	Dr. NTR CAE, Bapatla	Dr. NTR CAE, Bapatla	ANGRAU	Around 40 participants from faculty,

	Profitability of Agriculture"						farmers
2022	Kisan Mela – Exhibition of latest farm machinery and post-harvest technologies	26.03. 2022	26.03. 2022	Dr. NTR CAE, Bapatla	Dr. NTR CAE, Bapatla	ANGRAU	Around 600 participants from Farmers, Officials from Agril dept.
2023	Application of AI and ML in Agricultural Engineering – Unleashing the Potential for Food Security	04.09. 2023	05.09. 2023	Online	Dr. NTR CAE, Bapatla	NAHEP - IDP	Around 150 participants (Faculty, Ph.D. scholars, RAs etc.) from SAUs
2023	Innovations in Agri Robotics	30.12. 2023	30.12. 2023	Dr. NTR CAE, Bapatla	Dr. NTR CAE, Bapatla	NAHEP - IDP	Around 50 participants from faculty, farmers, Industrialists
2024	Interaction Meeting – Agri Machinery / Equipment Manufacturing Industries of Andhra Pradesh	03.12. 2024	03.12. 2024	Dr. NTR CAE, Bapatla	Dr. NTR CAE, Bapatla	ICAR and ANGRAU	Around 30 participants from faculty, farmers, Industrialists



Fig. 12. Dr. R. Sarada Jaya Lakshmidevi, Honorable Vice Chancellor of ANGRAU, Lam, Guntur delivering inaugural message for the programme on "Application of AI and ML in Agricultural Engineering – Unleashing the Potential for Food Security"



Fig. 13. Dr. Manoj Karkee, Professor, Dept. of Biological Systems Engineering, Washington State University, USA delivering lecture "Application of AI and ML in Agricultural Engineering – Unleashing the Potential for Food Security"

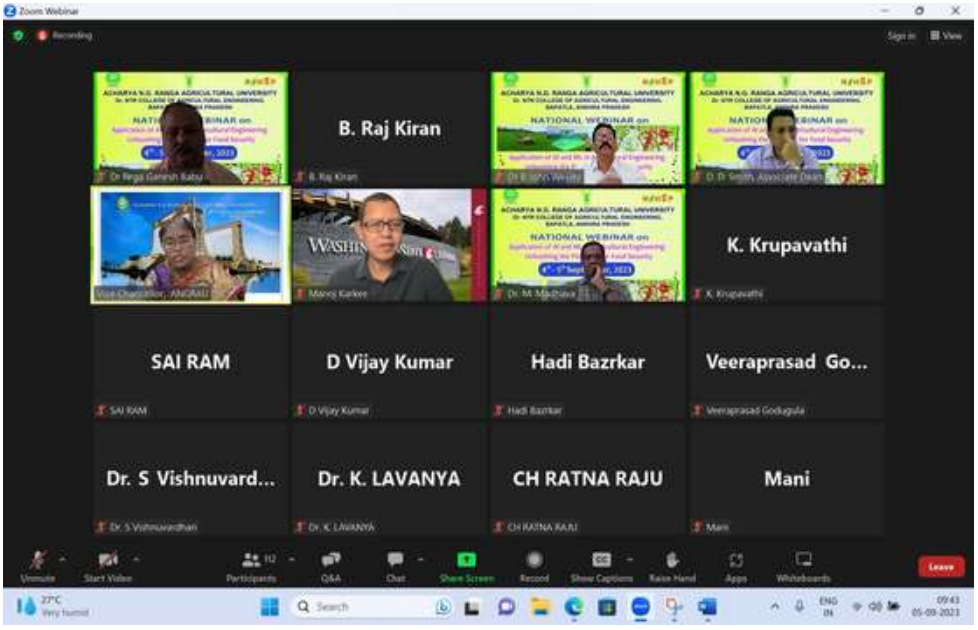


Fig. 14. Participant attending programme on "Application of AI and ML in Agricultural Engineering – Unleashing the Potential for Food Security" through online Zoom platform



Fig. 15. Inauguration program on "Innovations in Agri Robotics" at Dr. NTR College of Agricultural Engineering, Bapatla on 30.12.2023



Fig. 16. Participants and students attended for "Innovations in Agri Robotics" at Dr. NTR College of Agricultural Engineering, Bapatla on 30.12.2023



Fig. 17. Programme on "Interaction Meeting – Agri Machinery / Equipment Manufacturing Industries of Andhra Pradesh" at Dr. NTR College of Agricultural Engineering, Bapatla during 2024-25

Table 11. Year wise total number of Publications by the students

Year	Number of Publications					
	Research Papers	Books	Manuals	Study Materials	Conference Papers	Total
2020	4					**4**
2021	5					**5**
2022	6					**6**
2023	11				2	**13**
2024	10					**10**
2025	4					**4**
Total	**40**				**2**	**42**

Table 12. List of Research Papers Published by Faculty

Sl. No.	Authors	Year	Title of the paper	Name of the Journal	Vol. and page numbers
1	T.Mahesh Babu, **A.Ashok Kumar**, **K.V.S.Rami Reddy** and H.V.Hema Kumar	2020	Development of Passive-Passive Combination Tillage Implements Suitable for Mini-tractor	Current Journal of Applied Science and Technology	39 (24): 1-12
2	M. Lakshmi Durga, **A. Srinivasa Rao**, and **A. Ashok Kumar**	2020	Performance Evaluation of Single Row Low Horse Power tractor operated vegetable transplanter	Current Journal of Applied Science and Technology	39(44):37-44
3	**Arudra Srinivasa Rao**, Tapan Kumar Khura, Roaf, Indra mani, H.L Kushwaha	2020	Optimization of Operational Parameters for Tractor Operated Swinging lance sprayer by using ASPEE –SPGF	Current Journal of Applied Science and Technology	39(48):495-502
4	G. Gayathri, **D. Vijay Kumar**, K. Rupavathi, M.V. Ramana and K.V. Harshanjali	2020	Development of a mini groundnut kernal grader	Indian Journal of Ecology	47 Special (11): 262-265
5	Ajay, A., **Rami Reddy, K.V.S. and Ashok Kumar, A**.	2021	Study of Physical Properties of Zea mays in the Development of Seed Metering Unit	International Journal of Agriculture, Environment and	14 (2): 159-163

				Biotechnology	
6	**Arudra Srinivasa Rao**, Tapan Kumar Khura, Roaf, Indra mani, H.L Kushwaha	2021	Development evaluation of tractor operated swinging lance sprayer for field crops	The Pharma Innovation	10 (6):18-23
7	**Arudra Srinivasa Rao**, Tapan Kumar Khura, Roaf, Indra mani, H.L Kushwaha	2021	Spray droplet deposition, collection and analysis techniques; A review	The Pharma Innovation	10 (6): 156-165.
8	**B Raj Kiran** B Sanjeeva Reddy C Ramana HV Hema Kumar, Y Bindiya B Hari Babu	2021	Selection of dibble radius in automatic dibble type vegetable transplanter based on zero velocity condition in planting	The Pharma Innovation Journal	10 (5), 1507-1513
9	B. Krishna Kanth **C. Ramana** **S. Joseph Reddy** L.Edukondalu B. Ravindra Reddy	2021	Selection of Suitable Type of Nozzles for Development of Electronic Induction Nozzle	Current journal of Applied Science and Technology	40 (39), 1-10
10	Anitha.G, **Ashok Kumar.A,** **Srinivasa Rao.A**, and Ch. Someswara Rao	2022	Mini Tractor – mounted Sensor-based Aqua Groundnut Planter	Agricultural Mechanization in Asia, Africa, and Latin America (AMA)	53(1), 21-26
11	Anitha.G, **K.V.S. Rami Reddy.** **A, Ashok Kumar** Dr. L. Edukondalu and V. Srinivasa Rao	2022	Physical and mechanical properties of cotton seeds	The Pharma Innovation Journal	Special 11 (10), 1331-1333
12	B. Nageswar **B. Hari Babu** **C. Ramana** K. Madhusudhan Reddy B. Sreenivasula Reddy	2022	Studies on soil and groundnut plant parameters for design and development of a groundnut digger-Cum-Stripper	The Pharma Innovation Journal	11 (12): 3034-3038
13	K.T. Arya **C. Ramana** B. Ravindra Reddy	2022	Comparison of Anthropometric Parameters of	Current journal of Applied Science and	41 (44): 53-60

	B. Hari Babu M.V. Ramana		Women and Men Agricultural Workers Pertaining to Tractor Operator Workspace	Technology	
14	B Nageswar, **B Hari Babu, C Ramana,** K Madhusudhana Reddy and B Sreenivasula Reddy	2022	Studies on soil and groundnut plant parameters for design and development of a groundnut digger-cum-stripper	The Pharma Innovation Journal	11(12): 3034-3038
15	**D Vijay Kumar, C Ramana, K V S Rami Reddy, S** Kaleemullah, B Ravindra Reddy and S Pratibha Sree	2022	Studies on Different Methods of Sowing for Blackgram (*Vigna mungo* L.) under Rice Fallows	The Andhra Agricultural Journal	70 (1): 135-143
16	**D. Vijay Kumar C. Ramana K.V.S. Rami Reddy** S. Kaleemullah Ravindra Reddy	2023	Optimization of Machine and Operational Parameters in the Development of Stubble Manager cum Crop Planter	International Journal of Plant & Soil Science	35(16): 12-23
17	Manojkumara, **Srinivasarao, A**., **B. Rajkiran** Krupavathi, K., AshokKumar. A & Reddy, K. V. S. R.	2023	Development and Optimization of Operational Parameters for Tractor-Operated Boom Sprayer for Field Crops	International Journal of Environment and Climate Change	13 (10): 1127–1134
18	Venkatesh, K. **Kiran, B. R**., **Rao, A. S**., & Kumar, K. N.	2023	Development of Dibble Mechanism for Tractor Operated Two Row Semi–Automatic Vegetable Transplanter	Asian Journal of Agricultural Extension, Economics & Sociology	41 (12): 221-228
19	B Raj Kiran, B Sanjeeva Reddy, C Ramana, HV Hema Kumar and Y Bindiya	2023	Development of plant carrier type conveying unit for paper pot vegetable seedlings	The Pharma Innovation	12 (4): 2558-2565
20	S. Sai Mohan, C. Ramana,	2023	Investigating physical and	Journal of cotton research	37 (1): 153-158

	A. Ashok Kumar, K. Madhusudana Reddy		biometric properties of local grown cotton varieties during kapas picking stage to estimate the crop canopy characteristics	and development	
21	Ambiripuri Rahul A. Ashok Kumar K.V.S. Rami Reddy L. Edukondalu	2023	Design and Development of Chilli Planter for Direct Sowing of Seeds	Agricultural Mechanization in Asia, Africa and Latin America	54 (1)
22	Regatti Venkat, KVS Rami Reddy, A Ashok Kumar, R. Ganesh Babu Ch. Someswara Rao	2023	Investigation of Physical Properties of Turmeric Rhizomes About to Design a Metering Mechanism for Turmeric planting	Current Journal of Applied Science & Technology	42 (43): 45-52
23	**Veeraprasad** G,Srinavas I, **Rami Reddy K.V.S.**,Hemakumar H.V.and Srinivasa Rao V.	2023	Comparative Performance Analysis of a Remote-Controlled Paddy Weeder in Varied Soil Conditions	International Journal of Environment and Climate Change	13: 3515-3520
24	**Veeraprasad G.**, Srininvas I., **Rami Reddy K. V. S.**, Hemakumar H. V. and Srinivasa Rao V.	2023	Structural Analysis of the Differential Chassis for Development of Paddy Weeder Using Finite Element Analysis	International Journal of Plant & Soil Science	35 (22): 549-555
25	**Veeraprasad G.,** Srininvas I, **Rami Reddy KVS,** Hema Kumar HV and Srinivasa Rao V	2023	Structural analysis of the weeding attaching frame of robotic paddy weeder using finite element analysis	The Pharma Innovation Journal	12(11): 2164-2109
26	**Srinivasarao, A.**, T. K. Khura, R. A. Parray, I. Mani, H. L. Kushwaha, and S. K. Sarkar.	2023	Optimization of operational parameters for tractor operated swinging lance sprayer using	Agricultural Engineering International: CIGR Journal,	25 (3):109-120.

			Taguchi design.		
27	Malathi, K. M., **Rao, A. S**., **Kiran, B. R.**, &Babu, R. G.	2024	Study on Physical Properties of Groundnut Cultivated in Andhra Pradesh for Design of Remote-Operated Groundnut Planter.	Journal of Scientific Research and Reports	30 (7): 23-32
28	Koteswararao. K, **Rajkiran. B, Srinivasarao. A**, and Lavanya. K.	2024	Development of ExG, ExR, ExGR, HSV, CIELAB Images from RGB images using image segmentation Algorithm in Computer Vision Based Herbicite Spraying Applications	Journal of Scientific Research and Reports	30 (10): 501-508.
29	M. Vinayak, B. Hari Babu , A. Ashok Kumar, M. Madhava N. Ratna Kumari and C. Ramana	2024	Study the Physical Properties of Fertilizers and Plant Basin Parameters to Design the Tractor Operated Round Basin Maker cum Fertilizer Applicator for Orchard Crop	Journal of Scientific Research and Reports	30 (5): 487- 493
30	Maheswari, A. Ashok Kumar, B. Hari Babu Ch. Someswara Rao	2024	Study of the Feasibility of Happy seeder for Direct sowing of pulses in rice fallow fields of southern India	Agricultural Mechanization in Asia, Africa and Latin America	54 (1), 21-26
31	S. Rahaman **B. Hari Babu** **A. Ashok Kumar** K. Madhusudhana Reddy V. Srinivasa Rao	2024	Design of a Pendulum Impact Cutter to Measure Plant Stalk Cutting Force and Energy	Journal of Scientific Research and Reports	30(1): 51-62
32	S. Rahaman **B. Hari Babu** **A. Ashok Kumar** K. Madhusudhana Reddy V. Srinivasa Rao	2024	Economic Analysis and Feasibility of Tractor Operated Pulse Crop Harvester	Economic Affairs	69 (01): 509-516

33	A. Ajay**, K.V.S. Rami Reddy, A. Ashok Kumar** and Ch. Someswar	2024	Performance evaluation of twin-row planter for maize crop	Journal of Eco-friendly Agriculture	19 (1): 211-215
34	A. Ajay, **K.V.S.Rami Reddy**	2024	Economic Evolution of Twin Row Maize Planter: A Cost-Benefit Analysis	Environment and Ecology	42(1): 34-38
35	A. Ajay, **K.V.S. Rami Reddy, A. Ashok Kumar** and Ch. Someswar	2024	Performance evaluation of twin-row planter for maize crop	Journal of Eco-friendly Agriculture	19: 211-215
36	**A.Srinivasarao,** S.Mounika, M.saisudha, B.Revathi, T.Sravanthi and H.V.Hema Kumar.	2024	Development of Low-cost automatic sowing depth measurement unit for groundnut planter	Agricultural Mechanization in Asia, Africa, and Latin America (AMA)	54(4):22-28
37	**S. Sai Mohan, C. Ramana, A. Ashok Kumar, K. Madhusudana Reddy N. Sunitha**	2025	Energy use Efficiency and Economic Feasibility of Self-propelled Single Row Cotton Picker over Traditional Cotton Picking Practices	Indian Journal of Agricultural Research	59 Special: 162-169
38	Regatti Venkat, **KVS Rami Reddy, A Ashok Kumar,** K Krupavathi Ch. Someswara Rao	2025	Development and evaluation of laboratory turmeric seed rhizome metering mechanism	International Journal of Horticulture and Food Science	7 (9): 22-32
39	N. Prahlada, **G. Veeraprasad , K. V. S. Rami Reddy** and N. Ashok Kumar	2025	Study on Physico Engineering Properties of Cotton Seed for Designing of Metering Mechanism of Pneumatic Planter	Journal of Scientific Research and Reports	31 (6): 729-739
40	Ramavath Eliya, D Vijay Kumar, Raj Kiran Bakka, K Lavanya	2025	Design and Field Evaluation of an Automatic Inter and Intra-row Weeder for Wide-spaced Crops	Journal of Scientific Research and Reports	31 (6): 1053-1068

Table 13. List of Conference Papers Presented by Faculty

Sl. No.	Authors	Year	Title of the paper	Conference Details	Paper Number/Page Numbers
1.	G. Sindhuja**, A. Ashok Kumar, A. Srinivasa Rao,** Ch.Someswara Rao, **K.V.S Rami Reddy**	2023	Development of micro-controller-based roto planter for direct sowing of paddy suitable for mini tractor	57TH ISAE Convention, 06.11.2023 to 08.11.2023, UAS, Raichur	101
2.	B. Nagaraj, **A. Ashok Kumar, K.V.S Rami Reddy,** L. Edukondalu	2023	Remote-control operated smart sprayer suitable for field crops	57TH ISAE Convention, 06.11.2023 to 08.11.2023, UAS, Raichur	101

Table 14. List of Guest Lectures Organized by Dept. of FMPE During Last Five Years

S. No.	Name of the Expert	Topic	Date
1	Dr. G.R.K. Murthy Principal Scientist NAARAM, Rajendranagar Hyderabad	1) Recent Trends and concept In-precision Agriculture	07.02.2020
		2) Frontiers in Agricultural Engineering way forward	
		3) Technology in Education	
2	M/s. Nano Scientific Research Centre Pvt. Ltd. 6th floor Siri Estates, R.S. Brothers, Ameerpet, Hyderabad	1) Introduction to various Robotic (5 Hours)	24.02.2021
		2) Laws of Robotics	25.02.2021
		3) Aurdino Programme	26.02.2021
		4)Robotic implementation	27.02.2021
		5) Sensor interfaces	28.02.2021
3	Dr. B. Sanjeeva Reddy Principal Scientist CRIDA Hyderabad	1) Machine Vision Technology in Agriculture	09.05.2022
		2) Unmanned Aerial Systems and Payload Technologies for Precession Agriculture	
4	Dr. P. Rajaiah Principal Scientist & Univ. Head (FMPE) Rajendranagar, Hyderabad	1) Agricultural robotics	21.05.2022
		2) Role of IOT Technology in Agriculture	
5	Dr. T. Senthil Kumar Principal Scientist (FMPE) Regional Centre, CIAE, Bhopal	Mechanizing Millet cultivation for higher productivity, profitability and sustainability	16.02.2023
6	Dr. P. Rajaiah Principal Scientist &	1) Mechanization needs of small holding farms of rain fed ecosystem	10.11.2023

	Univ. Head (FMPE) Rajendranagar, Hyderabad	2) Appropriate mechanization for conservation Agriculture	
		3) Automation of need based Agricultural operations	
7	Dr. Debaraj Behera Ex. Professor (FMPE) OUAT, Bhubaneswar	1) Agricultural Machinery for conservation Agriculture	12.12.2023
		2)Precession Agriculture Tools and Techniques	
		3) Smart Agriculture commercial Approach	
8	Dr. T. Senthil Kumar Principal Scientist (FMPE) Regional Centre, CIAE, Bhopal	Robotics and Automation in Agricultural Machinery	12.09.2025

Learning Resources

M.Tech. students are trained in various computer software, namely, AutoCAD, ProE, SolidWorks, MATLAB, Python, etc. during practical classes. Educational videos are played for the students for better understanding of the subject in various courses.

College Library (Digital)

Libraries and information services play a vital role in supporting high-quality education and learning environment amongst students, faculty members and post-graduate scholars. This College is having a separate block for Library. The library has a good collection of 12,190 subject matter books, 470 gift books, 56 NAHEP-IDP books and 461 project reports. Apart from these books, there are 97 books and 147 books exclusively for SC and ST students. To facilitate the students writing competitive examinations, library is enriched with GRE, GMAT, TOEFL, CAT, general studies, computer books with CDs and different dictionaries (Telugu to English and English to English), handbooks, annual reports, world atlas, etc. Apart from them, there are a wide variety of Telugu and English daily newspapers, popular weekly and monthly magazines like *Rythunestham*, *Annadata*, etc. and competitive magazines. They are being kept for students' general reading.



Fig. 18. Reading facility



Fig. 19. Magazine Section



Fig. 20. Digital UG Computer Laboratory



Fig. 21. PG Students Computer Laboratory

The issues and returns of the library books are strictly on library card basis. The College Library is equipped with 2 good Xerox machines and 2 computers along with printers. Apart from them, 32 computers with internet facility are being maintained under digital libraries. The entire library is equipped with 09 air conditioners for comfortable study of students. Computers having high end configuration with 40 MBPS Internet speed to utilize the electronic information, namely CeRA, CAB Abstracts, KrishiKosh, EBSCO, Indiastat, CRC Netbase e- books, CAB e-books, Elsevier e-books, CMIE, EPWRF India Times Series, EZ Proxy, DELNET, OPAC, etc. for the improvement of academic and research standards. The library is completely under survailence with 14 no.s of CC cameras monitoring all the time.

Number of books: 12190

Periodicals: 517

Student Development

College is imparting very good education to the students of M.Tech.(Farm Machinery and Power Engineering) for their overall academic development.

Student Intake and Attrition

College has been offering M. Tech. (Agricultural Engineering) in the Department of Farm Machinery and Power Engineering from 2006-07 onwards. So far, 41 M.Tech.(Farm Machinery and Power Engineering) students have been awarded their degrees. The college has 2 M.Tech.(Farm Machinery and Power Engineering) graduates on roll during the academic year 2025-26.

Table 15. Student Intake and Attrition for the College in M.Tech. (Farm Machinery and Power Engineering) Programmes

Name of Degree Programme	Actual student admitted in last five years						Attrition (%)					
	Y 1	Y2	Y3	Y4	Y5	Y6 *	Y1	Y2	Y3	Y4	Y5	Y6 *
M.Tech. – Farm Machinery & Power Engineering	1	3	2	2	0	2	0	0	0	0	0	0

** Current year 2025-26*

Table 16. Average Number of Students in Theory and Practical Classes

Sl. No.	Name of the Degree Programme	Batch of Students in Theory Class	Batch of Students in Practical Class
1	M.Tech. (Farm Machinery and Power Engineering)	2	2

Admission Process

Registration of the courses is being done in the College during Orientation Session for M.Tech. & Ph.D. programmes separately. Students must fill in the registration material and the

registration slips will be passed on to the respective courses in charge. Academic Calendar is scrupulously prepared by the University for M.Tech. & Ph.D. programmes separately.

M. Tech. (Agricultural Engineering)

Admission process carried out to the M.Tech. programme in Farm Machinery and Power Engineering is through entrance examination (JRF) conducted by the ICAR every year (since 2018-19) as per the reservation policy of the state. Earlier ANGRAU conducted the entrance examination based on the B.Tech. (Agricultural Engineering) syllabus. Present intake is 5 students in the Farm Machinery and Power Engineering at Dr. N.T.R. College of Agricultural Engineering, Bapatla. Apart from regular admission, over and above, 15% of students secure admission under ICAR quota with ICAR fellowship. From the academic year 2017-18 onwards, provision is made for international students also for PG programmes.

Examinations and Evaluation Process

Examinations for all the courses are being conducted with fool proof and transparency. Most of the courses have theory as well as practical components. Each course is evaluated for 50 marks.

M. Tech. (Agricultural Engineering)

Mid-semester examinations are held after 50% of the semester working days in half the syllabus, whereas semester final theory examinations are conducted after completion of the semester. Mid-Semester examinations and semester final theory examinations are conducted for 30 and 100 marks, respectively. Assignments (including term papers) are evaluated for 20 marks.

Mid-semester examination question papers are set by the courses in-charge itself and conduct the examination. Question paper has two parts (Part-A and Part-B). Part-A contains bits in the form of filling in the blanks, multiple choice questions, matching, formulae, statements, one sentence answers, etc. for 30 marks. Part-B contains 4 questions of essay type, derivations, problems solving, etc. for 30 marks. After examination is completed, the concerned course in-charge evaluates the answer scripts and shows the students for verification within 10 days.

Practical final examination is being conducted in the laboratory, workshop, computer centre, drawing hall or examination hall depending upon the nature of the course. Practical

components are evaluated on 50:50 basis, i.e., 50% of the marks for internal evaluation based on record work, experimentation, slip tests, quizzes, attendance, etc. and the remaining 50% of the marks are for final practical examination. Those students who obtained more than 75% of attendance are allowed to appear for both the practical final and semester final theory examinations.

Question papers for the final semester theory examinations are also set by the courses concerned in-charge and sent to the University. Question paper has two parts (Part-A and Part-B). Part-A contains short answer questions for 40 marks. Out of 10 questions, students must attempt 8 questions. Part-B contains 8 questions of essay type, derivations, problems solving, etc. for 60 marks, but students have to attempt any 5 questions. All these question papers are moderated by senior faculty at University level. Then, moderated question paper is printed and supply the sealed bundles of the question papers to the concerned Colleges one day before the start of the semester final theory examinations. They are put under lock and key by the Associate Dean of the College. On the day of examination, the concerned question paper bundle is opened 15 min prior to the scheduled start of the examination. Examinations are conducted with the help of course in-charge and departmental staff. The examination is held for 3 hours. After examination, all the answer scripts are sealed and put under custody. All the bundles of the answer scripts are sent to university. There, answer scripts are jumbled/randomized and coded. Course in-charge, who dealt with the same course, is assigned the evaluation duty by Dean of Post Graduate Studies. Generally, each teacher is allowed to evaluate about 30 answer scripts in a day. After evaluation is completed, the answer scripts are decoded by university, and the results are sent to the College. The course in-charge prepares the Performance Register as per the weightage and declares the results. Those students who obtained more than 75% of attendance are permitted to appear for the semester final theory examination. Students must score 50% of marks to pass both semester theory examination and practical final examination. Also, they need to obtain 6.0 Grade Point to clear the course.

Comprehensive written and oral examinations are conducted after the course work is completed. Written examinations in core courses only are being held for M.Tech. students. Students must obtain 50% in written examination. Oral examination for these students is being conducted by inviting External Examiner from other SAUs or IITs.

Personality Development

College is continuously striving hard to overall development of all the students. Several personality development classes are being conducted regularly apart. Special classes also being conducted for backlog students and SC-ST students. Much emphasis also being given to improving the spoken English to perform better in placement interviews.

Research Contingency

ANGRAU is in its budget allocation, PG research contingency fund is allocated annually to support the M.Tech. and Ph.D. students in the department of Farm Machinery and Power Engineering (FMPE).

Table 17. FMPE Contingency Grant during Last Five Years

S. No.	Year	Grand received (Rs.)
1	2020-21	165000
2	2021-22	407000
3	2022-23	418000
4	2023-24	330000
5	2024-25	200000
6	2025-26	--

Outcome/output

Various M. Tech research works are been completed in the department of Farm Machinery and Power Engineering for the last five year are: (1) Development and Evaluation of Twin-row Maize Planter (2) Development and Evaluation of Semi-Automatic Inter and Intra Row Weeder for Wider Row Spaced Crops (3) Development and Evaluation of Tractor drawn cotton stalk up rooter cum shredder (4) Development of Sensor based paddy planter for automatic seeding (5) Development of remote control operated sensor based site specific chemical applicator (6) Development of high clearance self-Profelled Chilli Harvester (7) Development and evaluation of multipurpose Equipment for row crops (8) Design and Development of a Microcontroller –Based Variable Rate Sprayer for Field Crops (9) Development of Micro-Controller Based Roto Planter For Direct Sowing of Paddy Suitable for Mini Tractor (10) Development and Performance Evaluation of Tractor Operated Two Row

Semi-Automatic Vegetable Transplanter (11) A Real Time Vision Based Inter Row Weed Detection and Herbicide Spraying System (12) Development of Remote –Control Operated Self-Propelled Groundnut Planter (13) Development and Performance Evaluation of Automatic Inter and Intra row Weeder for Wide Spaced Crops, and (14) Design and Development of Robotic Pneumatic Planter for Cotton, which were evaluated and field tested.

Table 18. Awards and Honors of the Faculty during Last Five Years

Sl. No.	Name	Awards and Honors
1.	Dr. A. Srinivasa Rao, Assistant Professor	Meritorious Teacher award-2023 received from the ANGRAU
2.	Er. N. Kishore Assistant Professor	Best Agri Engineer Award from ICFRE–HFRI, Shimla, in association with the National Medicinal Plant Board, RRS, Himachal Pradesh (Govt. of India) and ISAHARD Society, New Delhi, during 13th International Conference
3.	Er. N. Kishore Assistant Professor	Netaji Subhas- ICAR International Fellowship – 2025 by Agricultural Education Division, Indian Council of Agricultural Research, New Delhi

Externally funded projects of Faculty

1. Dr. B. Raj Kiran, Assistant Professor and Principal Investigator for project entitled “Skill Development on operation, maintenance of advanced farm machinery and allied knowledge for making the Agro - entrepreneurs / job seeking opportunities for unemployed “during the year 2023-24 under ICAR-SCSP Project with Rs.23.78 lakhs budget.

6.4.11 Certificate

I, the Associate Dean of the Dr. NTR College of Agricultural Engineering, Bapatla, Andhra Pradesh hereby certifies that the information contained in Sections 6.4.1 to 6.4.9 are furnished as per the records available in the college and degree awarding university.

Signature of the Associate Dean with Date and Seal